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MILITARY SPECIFICATION

CARTRIDGE, EXPLOSIVE BOLT, LAU-6/A

1. SCOPE

1.1 Scope. This specification establishes the requirements for one type of Cartridge, Explosive Bolt designated as LAU-6/A hereafter referred to as the charge, for use in conjunction with the AIR-2A Air-to-Air Rocket, and the ATR-2A Air-to-Air Practice Firing Rocket.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition

PUBLICATIONS

Air Force

AFM 127-100	Explosives Safety Manual

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DRAWINGS

Douglas Aircraft Company, Inc.

2602441	Plug, Assembly, Shorting Charge
3602079	Cartridge, Explosive Bolt LAU-6/A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provisions for preproduction testing (see 6.4.a).

3.2 Components. The charge shall consist of the following major components:

a. Charge, Assembly
b. Plug, Assembly, Shorting, Charge

3.2.1 Charge assembly. The charge assembly shall be constructed in conformance to Douglas Aircraft Company Drawing No. 3602079 and its subsidiary drawings.

3.2.1.1 Serviceable life. The serviceable life of the charge shall be a minimum of eighteen months.

3.2.2 Plug, assembly, shorting, charge. The shorting plug shall be constructed in conformance with Douglas Aircraft Company Drawing 2602441.

3.2.3 Interchangeability. All functionally and dimensionally interchangeable parts shall have the same manufacturer's part number.

3.3 Safety regulations. All safety precautions required by the following regulation shall be taken in the manufacture, testing, handling, storage and transportation of the charge.

a. Air Force, AFM 127-100, Explosive Safety Manual

3.4 Operational requirements

3.4.1 Electrical requirements

3.4.1.1 Continuity. With pin A grounded to the case, the resistance between each of the other pins and the case shall be 9 $\pm$ 3 ohms.

WARNING

No more than 10 milliamperes of current shall be applied to the detonator circuit except when firing of the charge is intended.

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3.4.1.2 Leakage resistance. The leakage resistance between each connector pin and the case shall be greater than 0.5 megohm at 500 volts direct current.

3.4.1.3 Plug, shorting. The shorting plug shall, when attached to the charge, interconnect the pins of the detonator circuit in such a manner as to prevent inadvertent introduction of current to the detonator bridge wires as determined by the test described in "4.5 TEST METHODS".

3.4.1.4 Power. The charge shall function satisfactorily when a voltage of 28 volts direct current or the electrical power of the aircraft pulse generator is applied as controlled by the current limiting resistor in the aircraft wiring system.

3.4.1.5 Current. Current criteria of the charge shall be as follows:

a.	Maximum test current	10 milliamperes
b.	Maximum no-fire current	45 milliamperes
c.	Nominal firing current	1 ampere

These criteria apply to each individual detonator circuit.

3.5 Environmental requirements. The charge shall be capable of satisfactory operation after being subjected to the following environmental conditions:

a. High temperature
b. Low temperature
c. Vibration
d. Drop
e. Humidity
f. Salt Spray

3.6 Identification of product. The charge shall be marked for identification in accordance with MIL-STD-130.

3.6.1 Identification of lots. Each production lot as defined in MIL-STD-1168 shall be identified and numbered in accordance with that specification or as specified by the procuring activity, and shall be recorded on Ammunition Data Cards.

3.6.2 Ammunition data cards. Ammunition Data Cards shall be initiated, maintained and distributed in accordance with MIL-STD-1167.

3.7 Workmanship. The charge shall be fabricated and finished in such a manner that criteria of appearance, fit, and adherence to specified tolerances are observed. Particular attention shall be given to neatness and thoroughness of marking of parts, plating, painting, potting, and freedom of parts from burrs and sharp edges that may result in injury to personnel or damage to adjacent equipment.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. Quality assurance provisions shall be in accordance with the applicable component detail specifications.

4.1.2 Packing and marking. The inspector shall ascertain that the packing and marking of the canisters conform to this specification.

4.2 Preproduction testing.

4.2.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production charge. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2 Preproduction test report. In the event that preproduction testing is conducted at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity. No production deliveries shall be made prior to acceptance based on the preproduction tests and preproduction test report.

4.2.3 Preproduction tests. Preproduction tests shall consist of all tests specified in this specification (see 6.4.a).

4.3 Acceptance tests. Acceptance tests (see 6.4.b) of the charge shall consist of the following:

a. Individual Tests
b. Sampling Tests

4.3.1 Individual tests. Each charge shall be subjected to the following tests as described under "4.5 TEST METHODS".

a. Examination of Product
b. Conformance to Drawings
c. Electrical Tests

4.3.2 Sampling plan. Five percent of each production lot, or twenty charges, whichever is greater, shall be selected for subjection to the sampling test specified in this specification.

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4.3.2.1 Sampling tests. Sampling tests shall consist of the following:

a. Functional Test
b. High Temperature
c. Low Temperature
d. Vibration Test
e. Drop Test
f. Humidity Test

Sampling plan samples shall have withstood the individual tests.

4.3.2.2 Rejection and retest. When one item selected from a production lot fails to meet the specified tests, no items still on hand or later produced shall be accepted until the extent and cause of failure is determined.

4.3.2.3 Individual tests may continue. For operational reasons, individual tests may be continued pending investigation of a sampling test failure. But final acceptance of charges on hand or produced later shall not be made until it is determined that items meet all the requirements of this specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in charges already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions. Test conditions shall be as specified in each test in "4.5 TEST METHODS".

4.5 Test methods

4.5.1 Examination of product. Each charge shall be examined to determine that the requirements for workmanship are fulfilled.

4.5.2 Conformance to drawings. Each charge shall be inspected for strict conformance to the engineering drawings.

4.5.3 Continuity test. With pin A, Douglas Aircraft Company drawing 3602079, grounded to the case, measure the resistance between pins B and C and the case. The measured resistance between each pin and the case shall be 9 $\pm$ 3 ohms. Measured resistance not within these limits shall be cause for rejection. This test shall be repeated after each environmental test.

WARNING

The maximum current for this test shall be 10 milliamperes.

4.5.3.1 Leakage resistance. After removing the ground, established in the continuity test, from pin A, apply 500 volts direct current between each pin separately and the case. Measure the resistance between pin and case. Measured resistance of 0.5 megohm or less shall be cause for rejection.

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4.5.3.2 Continuity, shorting plug. Measure resistance between any one of the three sockets in the shorting plug and each of the other two separately. Failure to achieve an indicated closed circuit shall be cause for rejection.

4.5.4 Functional test. After withstanding the individual tests and all environmental tests except the drop test specified in this specification, the sample charges shall be mounted in a fixture simulating attachment to the Launcher, Short Rail, Type MA-5. Electrical power of 28 volts direct current at a rate of a minimum of 1 ampere shall be applied. Failure to fire high-order shall be cause for rejection of the lot. This test shall be conducted behind a suitable barricade.

4.5.5 Environmental tests. The environmental tests specified herein shall be in accordance with the procedures of MIL-E-5272 specified in each test, except the provisions for operation of equipment during testing shall not apply. Immediately after each environmental test, the sample charge shall be subjected to the continuity and leakage resistance tests. Failure to satisfy the continuity and leakage resistance tests shall be considered as failure to withstand the environmental test and shall be cause for rejection.

4.5.5.1 High temperature test. The charge shall be subjected to the high temperature test described in MIL-E-5272.

4.5.4.2 Low temperature test. The charge shall be subjected to the low temperature test described in procedure II, MIL-E-5272.

4.5.4.3 Vibration test. The charge shall be subjected to the vibration test described in procedure III, MIL-E-5272.

4.5.5.4 Drop test. Each sample charge shall be dropped from a height of 12 feet onto a surface conforming to the following:

a. Thickness - 0.50 inch, minimum.
b. Length - 1.5 times the greatest dimension of the charge.
c. Width - 1.5 times the greatest dimension of the charge.
d. Surface - Reasonably smooth and flat.
e. Hardness - Brinnell 203.
f. Support - 3 inches of concrete, minimum.

4.5.5.4.1 Drop attitudes. The sample charge shall be subjected to one impact in each of the following attitudes:

a. Pin end up.
b. Pin end down.
c. Longitudinal axis horizontal.
d. Longitudinal axis 45 degrees from vertical, pin end up.
e. Longitudinal axis 45 degrees from vertical, pin end down.

The charge shall not detonate on impact.

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4.5.5.4.2 Check tests. After being subjected to the procedures of 4.5.5.4.1, the charge shall be subjected to examination of product, the continuity test, and the leakage resistance test.

4.5.5.4.3 Rejection criteria. Detonation during drop test, damage or deformation of sufficient extent to impair handling safety, shall be considered failure to satisfy the drop test and cause for rejection.

4.5.5.5 Humidity test. The charge shall be subjected to the humidity test described in procedure I, MIL-E-5272.

4.5.5.6 Salt spray test. The charge shall be subjected to the salt spray test described in MIL-E-5272.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each charge shall be individually preserved and packaged with shorting plug installed in accordance with the requirements of method IA-8 of MIL-P-116.

5.1.2 Level C. Each charge shall be preserved and packaged in accordance with industry's best commercial practice to prevent deterioration and provide protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified by the procuring activity, three (3) charges shall be cushioned and immobilized in a container conforming to PPP-B-636.

5.2.2 Level C. Charges packaged as described in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the uniform freight classification rules or regulations of other carriers applicable to the mode of transportation. Packing shall also be accomplished in a manner which will provide adequate safety and protection against damage.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Nomenclature. The following marking shall be applied to the cartridge containers as designated in MIL-STD-129:

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6. NOTES

6.1 Intended use. The charge covered by this specification is intended to provide a capability to jettison the Rocket, Air-to-Air, Type AIR-2A and the Rocket, Practice Firing, Type ATR-2A by means other than conventional launching when mounted on the rail launcher.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Whether preproduction will be conducted by the procuring activity (see 4.2).
c. Levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.1 Preproduction test samples. The contract shall specify that a minimum of fifteen units representative of the production charges will be submitted to the procuring activity for test and evaluation. The contract shall specify the point of inspection for these tests.

6.3 Production delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions

a. Preproduction tests. Preproduction tests are those tests performed prior to production on representative models of the production charge to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction samples.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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